



Statistical Analysis

BAS331 – BAS332

Assignment 5

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TA:

Chapter 3

Test of Hypotheses for two parameters

Test for: $\mu_1 - \mu_2$, $\sigma_1^2 - \sigma_2^2$ and $p_1 - p_2$

1] A sample of 40 observations is selected from one population with a population standard deviation of 5. The sample mean is 102. A sample of 50 observations is selected from a second population with a population standard deviation of 6. The sample mean is 99.

(a) Test the hypothesis that $\mu_1 = \mu_2$ against the alternative that $\mu_1 \neq \mu_2$. Use $\alpha = 0.04$

(b) What is the p -value?

2] As part of a study of corporate employees, the director of human resources for PNC Inc. wants to compare the distance traveled to work by employees at its office in downtown Cincinnati with the distance for those in downtown Pittsburgh. A sample of 35 Cincinnati employees showed they travel a mean of 370 miles per month. A sample of 40 Pittsburgh employees showed they travel a mean of 380 miles per month. The population standard deviations for the Cincinnati and Pittsburgh employees are 30 and 26 miles, respectively.

(a) Test the hypothesis that there is significant difference in the mean number of miles traveled per month between Cincinnati and Pittsburgh employees. Use $\alpha = 0.05$

(b) What is the p -value?

3] Mary Jo Fitzpatrick is the vice president for Nursing Services at St. Luke's Memorial Hospital. Recently she noticed in the job postings for nurses that those who are unionized seem to offer higher wages. She decided to investigate and gathered the following information.

Group	Sample size:	Sample Mean Wage	Population Standard deviation:
Union	40	\$20.75	\$2.25
Nonunion	45	\$19.80	\$1.90

Would it be reasonable for her to conclude that union nurses earn more? Use the 0.02 significance level. What is the p -value?

4] A recent study compared the time spent together by single- and dual-earner couples. According to the records kept by the wives during the study, the mean amount of time spent together watching television among the single-earner couples was 61 minutes per day, with a standard deviation of 15.5 minutes. For the dual-earner couples, the mean number of minutes spent watching television was 48.4 minutes, with a standard deviation of 18.1 minutes. At the 0.1 significance level, can we conclude that the single-earner couples on average spend more time watching television together? There were 15 single-earner and 12 dual-earner couples studied.

5] The Tampa Bay (Florida) Area Chamber of Commerce wanted to know whether the mean weekly salary of nurses was larger than that of school teachers. To investigate, they collected the following information on the amounts earned last week by a sample of school teachers and a sample of nurses.

School Teachers (\$)	1095	1076	1077	1125	1034	1059	1052	1070
	1080	1092	1082	1079				
Nurses (\$)	1091	1140	1071	1021	1100	1109	1075	1079

Is it reasonable to conclude that the mean weekly salary of nurses is higher? Use the 0.1 significance level.

6] A random sample of 15 items from the first population showed a mean of 50 and a standard deviation of 5. A sample of 12 items for the second population showed a mean of 46 and a standard deviation of 15.

Test the hypothesis that $\mu_1 = \mu_2$ against the alternative that $\mu_1 \neq \mu_2$. Use $\alpha = 0.1$

7] A recent survey compared the costs of adoption through public and private agencies. For a sample of 16 adoptions through a public agency, the mean cost was \$21045, with a standard deviation of \$835. For a sample of 18 adoptions through a private agency, the mean cost was \$22840, with a standard deviation of \$1545.

Test the hypothesis that $\mu_1 = \mu_2$ against the alternative that $\mu_1 \neq \mu_2$. Use $\alpha = 0.1$

8] Suppose you are an expert on the fashion industry and wish to gather information to compare the amount earned per month by models featuring Liz Claiborne attire with those of Calvin Klein. The following table gives the amount (\$000) earned per month by a sample of 15 Claiborne models and a sample of 12 Klein models.

Claiborne models:	\$5.0	\$4.5	\$3.4	\$3.4	\$6.0	\$3.3	\$4.5	\$4.6
	\$3.5	\$5.2	\$4.8	\$4.4	\$4.6	\$3.6	\$5.0	
Klein models.	\$3.1	\$3.8	\$4.0	\$3.6	\$3.7	\$3.8	\$3.6	\$3.6
	\$4.9	\$5.9	\$2.3	\$4.0				

Is it reasonable to conclude that Claiborne models earn more? Use the 0.1 significance Level.

9] The management of Discount Furniture, a chain of discount furniture stores in the Northeast, designed an incentive plan for salespeople. To evaluate this innovative plan, 12 salespeople were selected at random, and their weekly incomes before and after the plan were recorded.

Salesperson	1	2	3	4	5	6	7	8	9	10	11	12
Before	320	290	421	510	210	402	625	560	360	431	506	506
After	340	285	475	510	210	500	631	560	365	431	525	619

Was there a significant increase in the typical salesperson's weekly income due to the innovative incentive plan? Use the .05 significance level.

10] The federal government recently granted funds for a special program designed to reduce crime in high-crime areas. A study of the results of the program in eight high-crime areas of Miami, Florida, yielded the following results.

	Number of Crimes by Area							
	A	B	C	D	E	F	G	H
Before	14	7	4	5	17	12	8	9
After	2	7	3	6	8	13	3	5

Has there been a decrease in the number of crimes since the inauguration of the program? Use the 0.01 significance level.

11] In a comparison study of two machines for manufacturing memory chips the following data were collected: Of 500 chips produced by machine I, 20 were defective; of 600 chips produced by machine II, 40 were defective.

(a) Can you conclude from these results that there is no significant difference between the proportions of defective items produced by the respective machines? (Use $\alpha = 0.05$)

(b) Compute and interpret the P-value.

12] Shingles is a virus infection of the nerves that causes a painful, blistering rash. It is the same virus that causes chickenpox. A new vaccine for shingles, Zostovax, was approved in May 2006 by the Food and Drug Administration (FDA). The approval was based on a large-scale clinical trial with 38500 people 60 or older, half of whom (group 1) received the vaccine and the other half (group 2) received a placebo. The vaccinated group had 315 cases of shingles; the placebo group had 642 cases. Let X_1 , X_2 denote the number of people in the vaccinated group and unvaccinated group, respectively, that contracted shingles, and denote the corresponding shingles prevalence rate for the two groups by p_1 , p_2 .

(a) Test the null hypothesis that the vaccine given to group 1 is not effective against the alternative hypothesis that it is. (Use $\alpha = 0.05$)

(b) Compute and interpret the P-value.

13] A firm has been experimenting with two different physical arrangements of its assembly line. It has been determined that both arrangements yield approximately the same average number of finished units per day. To obtain an arrangement that produces greater process control, you suggest that the arrangement with the smaller variance in the number of finished units produced per day be permanently adopted. Two independent random samples yield the results shown in Table (1).

- (a) Test the hypothesis that $\sigma_1^2 = \sigma_2^2$ against the alternative that $\sigma_1^2 \neq \sigma_2^2$ where, σ_1^2 and σ_2^2 are the variances of the number of finished units for the two assembly line arrangements respectively. Use $\alpha = 0.1$
- (b) Based on the result, in part (a) which of the two arrangements would you recommend?
- (c) Test the hypothesis that $\mu_1 = \mu_2$ against the alternative that $\mu_1 \neq \mu_2$ where, μ_1 and μ_2 are the means of the number of finished units for the two-assembly line arrangements respectively. Use $\alpha = 0.1$

Line 1	448	523	506	500	533	447	524	469	470	494	536	481	492
	567	492	457	497	483	533	408	453					
Line 2	372	446	537	592	536	487	592	605	550	489	461	500	430
	543	459	429	494	538	540	481	484	374	495	503	547	

Table (1): Number of Finished Units Produced per Day by Two Assembly Lines

14] Hippo grazing patterns in Kenya. In Kenya, human-induced land-use changes and excessive resource extraction has threatened the jungle ecosystem by reducing animal grazing areas and disrupting access to water sources. In Landscape & Ecology Engineering (Jan. 2013), researchers compared hippopotamus grazing patterns in two Kenyan areas — a national reserve and a community pastoral ranch. Each area was subdivided into plots of land. The plots were sampled (406 plots in the national reserve and 230 plots in the pastoral ranch) and the number of hippo trails from a water source was determined for each plot. Sample statistics are provided in the following table.

- (a) Can the researchers reliably conclude that the variability in number of hippo trails from a water source in the national reserve differs from the variability in number of hippo trails from a water source in the pastoral ranch? Use $\alpha = 0.1$. Use

$$f_{0.05}(405, 229) = 1 \text{ \& } f_{0.05}(229, 405) = 1$$

- (b) The researchers concluded that the mean number of hippo trails was higher in the national reserve than in the pastoral ranch. Do you agree? Use $\alpha = 0.1$

	National Reserve	Pastoral Ranch
Sample size:	406	230
Mean number of trails:	0.31	0.13
Standard deviation:	0.4	0.3

15] The following data are from Charles Darwin's study of cross- and self-fertilization. Pairs of seedlings of the same age, one produced by cross-fertilization and the other by self-fertilization, were grown together so that members of each pair were grown under nearly identical conditions. The data represent the final heights of the plants after a fixed period of time. Test the hypothesis that $\mu_1 = \mu_2$ against the alternative that $\mu_1 \neq \mu_2$. Use $\alpha = 0.05$.

(Note that: *The data are paired*)

Pair	Cross- fertilized	Self-fertilized
1	23.5	17.4
2	12.0	20.4
3	21.0	20.0
4	22.0	20.0
5	19.1	18.4
6	21.5	18.6
7	22.1	18.6
8	20.4	15.3
9	18.3	16.5
10	21.6	18.0
11	23.3	16.3
12	21.0	18.0
13	22.1	12.8
14	23.0	15.5
15	12.0	18.0

16] The compression strengths, measured in kgs., of cylinders produced by two different manufacturers were tested and the following data were recorded:

$$n_1 = 12, \bar{X}_1 = 240, S_1 = 10 \text{ \& } n_2 = 12, \bar{X}_2 = 210, S_2 = 7$$

(a) Test the hypothesis that $\sigma_1^2 = \sigma_2^2$ against the alternative that $\sigma_1^2 \neq \sigma_2^2$. Use $\alpha = 0.1$

(b) Test the hypothesis that $\mu_1 = \mu_2$ against the alternative that $\mu_1 \neq \mu_2$. Use $\alpha = 0.1$

17] Two random samples of sizes $n_1 = 11$ and $n_2 = 13$, respectively, were taken from two normal distributions $N(\mu_1, \sigma^2)$ and $N(\mu_2, \sigma^2)$, and the following data were obtained:

$\bar{X}_1 = 1.5, S_1 = 0.5$ & $\bar{X}_2 = 2.1, S_2 = 0.6$. Test the hypothesis that $\mu_1 = \mu_2$ against the alternative that $\mu_1 \neq \mu_2$. Use $\alpha = 0.1$

18] Two different weight loss programs are being compared to determine their effectiveness. Ten men were assigned to each program, so $n_1 = n_2 = 10$, and their weight losses are recorded below.

(a) Test the hypothesis that $\sigma_1^2 = \sigma_2^2$ against the alternative that $\sigma_1^2 \neq \sigma_2^2$. Use $\alpha = 0.1$

(b) Test the hypothesis that $\mu_1 = \mu_2$ against the alternative that $\mu_1 \neq \mu_2$. Use $\alpha = 0.1$.

Diet 1	3.4	10.9	2.8	7.8	0.9	5.2	2.5	10.5	7.1	7.5
Diet 2	11.9	13.1	11.6	6.8	6.8	8.8	12.5	8.6	17.5	10.3

19] A study published in Chemosphere reported the levels of the dioxin TCDD of 20 Massachusetts Vietnam veterans who were possibly exposed to Agent Orange. The amount of TCDD levels in plasma and in fat tissue were as follows:

<i>Veteran</i>	<i>TCDD levels In plasma</i>	<i>TCDD levels In fat tissue</i>
<i>1</i>	<i>2.5</i>	<i>4.9</i>
<i>2</i>	<i>3.1</i>	<i>5.9</i>
<i>3</i>	<i>2.1</i>	<i>4.4</i>
<i>4</i>	<i>3.5</i>	<i>6.9</i>
<i>5</i>	<i>3.1</i>	<i>7.0</i>
<i>6</i>	<i>1.8</i>	<i>4.2</i>
<i>7</i>	<i>6.0</i>	<i>10.0</i>
<i>8</i>	<i>3.0</i>	<i>5.5</i>
<i>9</i>	<i>36.0</i>	<i>41.0</i>
<i>10</i>	<i>4.7</i>	<i>4.4</i>
<i>11</i>	<i>6.9</i>	<i>7.0</i>
<i>12</i>	<i>3.3</i>	<i>2.9</i>
<i>13</i>	<i>4.6</i>	<i>4.6</i>
<i>14</i>	<i>1.6</i>	<i>1.4</i>
<i>15</i>	<i>7.2</i>	<i>7.7</i>
<i>16</i>	<i>1.8</i>	<i>1.1</i>
<i>17</i>	<i>20.0</i>	<i>11.0</i>
<i>18</i>	<i>2.0</i>	<i>2.5</i>
<i>19</i>	<i>2.5</i>	<i>2.3</i>
<i>20</i>	<i>4.1</i>	<i>2.5</i>

Test the hypothesis that $\mu_1 = \mu_2$ against the alternative that $\mu_1 \neq \mu_2$. where, μ_1 and μ_2 represent the true mean TCDD in plasma and in fat tissue, respectively. Assume the distribution of the differences to be approximately normal. Use $\alpha = 0.05$.

20] Two machines produce computer memory chips. Of 500 chips produced by machine 1, 20 were defective; of 600 chips produced by machine 2, 40 were defective.

(a) Test the hypothesis that $p_1 = p_2$ against the alternative that $p_1 \neq p_2$. Where, p_1 and p_2 are the proportions of defective chips produced by machine 1 and machine 2 respectively. Use $\alpha = 0.05$.

(b) Compute and interpret the P value.

21] To compare the effectiveness of two different methods of teaching arithmetic 100 students were divided into two groups of 50 students each. One group (the control group) was taught by traditional methods and the other group (the treatment group) by a new method. At the end of the term 65 students of the control group and 80 students of the treatment group passed the basic skills test.

(a) Test the hypothesis that $p_1 = p_2$ against the alternative that $p_1 \neq p_2$. where p_1 and p_2 are the true proportions of students who pass the basic skills test for the control and treatment groups, respectively. Use $\alpha = 0.05$.

(b) Compute and interpret the P value.

22] A random sample of 100 men and 100 women at a Southern college are asked if they have an automobile on campus. If 24 of the men and 13 of the women have cars,

(a) Test the hypothesis that $p_1 = p_2$ against the alternative that $p_1 \neq p_2$. where p_1 and p_2 are proportion of men and women who have cars on campus. respectively. Use $\alpha = 0.01$.

(b) Compute and interpret the P value.
